

NAME: _____

PERIOD: _____

DATE: _____

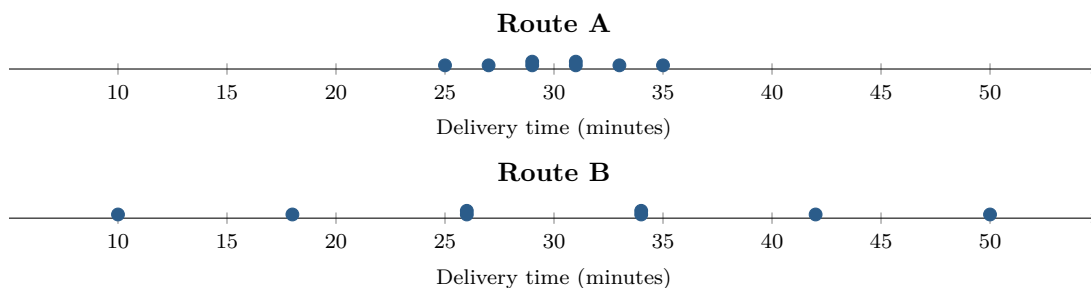
Same Mean, Different Deadline Risk

AP Statistics · Topic 4.8 (CED 2.9) · B: 2026-11-02 · E: 2026-11-23

[STAR] TODAY'S PROBLEM

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.



FIRST TAKE — before the video (5 min)

Which route would you assign to this customer? Cite one number or dotplot feature.

[BOOK] MEAN, STANDARD DEVIATION, AND SHIFTS (keep on the page)

For a discrete random variable, $\mu_X = \sum xP(X = x)$ and $\sigma_X = \sqrt{\sum (x - \mu_X)^2 P(X = x)}$. Interpret μ_X as the long-run average and σ_X as a typical distance from that mean, in context and with units. If $Y = X + c$, then $\mu_Y = \mu_X + c$ while $\sigma_Y = \sigma_X$: adding the same constant shifts every outcome but does not change spread. Match the parameter to the decision; equal centers do not imply equal tail risk.

Finish after the video:

DOK 1 (a) For each route, state its minimum and maximum modeled time and count the outcomes longer than 40 minutes.

DOK 2 (b) Verify the reported mean and population standard deviation for each route from the eight equally likely outcomes. Show the sums of squared deviations.

DOK 3 (c) Decide which route better matches this customer's priority. Cite the parameter and dotplot feature that settle your choice, and explain what the dispatcher's mean-only argument would mislead a reader into believing. Then suppose every delivery gets 5 minutes of loading time: state both transformed means and standard deviations and decide whether your route choice changes.

Frame: Route _____ fits the customer because $\sigma =$ _____ and the dotplot shows _____.

Frame: Adding 5 changes each mean to _____ but leaves the SDs at _____ and _____ because _____.

EXIT — turn this in

One thing the video changed about my first take: _____